

PROJECT II REPORT

Cluster-Based Analysis of the Glass Identification Data

MACT 4233 - Applied Multivariate Analysis

Dataset: Glass Identification Dataset

Methods: Hierarchical Clustering and K-means

Observations Used: 214 complete cases

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1 Project Overview

Clustering is an unsupervised learning technique that partitions observations into groups, called clusters, so that observations within the same group are more similar to one another than observations in other groups. Unlike classification or discriminant analysis, clustering does not use the class label during the grouping process; the goal is to discover the latent structure in the measurements themselves.

Clustering is useful whenever the researcher suspects that a population is heterogeneous but does not know the group boundaries in advance. In this project, the goal is to apply hierarchical clustering and K-means clustering to the GlassData dataset after withholding the Type variable. The resulting clusters are then compared with the known glass types using Hungarian-style label matching, because cluster labels are arbitrary and must be matched to the true labels before computing an error rate.

This report evaluates several clustering approaches on the selected Glass Identification dataset. Four hierarchical distance-linkage combinations are tested, K-means is evaluated using an elbow curve and repeated random starts, and all final partitions are compared using the Hungarian error rate and the R-squared measure based on within-cluster sum of squares.

2 Dataset and Variables

The selected dataset contains 214 complete observations and 11 variables. The observation unit is one glass sample. The class variable is Type, with six observed categories: Type 1, Type 2, Type 3, Type 5, Type 6, and Type 7. The class distribution is 70 samples from Type 1, 76 samples from Type 2, 17 samples from Type 3, 13 samples from Type 5, 9 samples from Type 6, and 29 samples from Type 7.

Table 1: Variables in the project dataset

Variable	Description	Type	Unit
Id	Sample identifier	Identifier	None
RI	Refractive index	Interval	Index
Na	Sodium measurement	Interval	Numerical
Mg	Magnesium measurement	Interval	Numerical
Al	Aluminum measurement	Interval	Numerical
Si	Silicon measurement	Interval	Numerical
K	Potassium measurement	Interval	Numerical
Ca	Calcium measurement	Interval	Numerical
Ba	Barium measurement	Interval	Numerical
Fe	Iron measurement	Interval	Numerical
Type	Glass type/class variable	Nominal	Category

For the clustering analyses, the Type label is withheld and only the nine continuous measurements are used: RI, Na, Mg, Al, Si, K, Ca, Ba, and Fe. The Id column is excluded because it is only an

identifier and does not contain measurement information. Since the variables are measured on different scales, all nine clustering variables are standardised to zero mean and unit variance before any distance calculation or K-means run is performed.

3 Hierarchical Clustering Analysis

3.1 Clustering setup and tested combinations

Hierarchical clustering builds a nested sequence of partitions represented as a dendrogram. At each step, the two closest groups are merged according to a chosen inter-cluster distance, called the linkage. Cutting the dendrogram at a chosen height gives a flat partition into k clusters. In this report, the dendrograms are cut at $k = 6$, matching the six observed glass types in the dataset.

Four distance-linkage combinations were evaluated:

- A: Euclidean distance with Ward.D2 linkage. Ward's criterion attempts to minimize the total within-cluster sum of squares at each merge step and usually produces compact clusters.
- B: Euclidean distance with Average linkage. Average linkage defines the distance between two clusters as the mean of all pairwise distances between observations in the two clusters.
- C: Manhattan distance with Ward.D2 linkage. Manhattan distance sums absolute differences across variables, while Ward.D2 keeps the compactness-oriented merging rule.
- D: Manhattan distance with Complete linkage. Complete linkage uses the maximum pairwise distance between members of two clusters and tends to form tight, well-separated groups.

3.2 Dendrogram output

Figures 1 through 4 show the dendrograms for the four combinations. In each figure, the red dashed horizontal line marks the cut at $k = 6$, and the resulting branches below that line define the clusters used for evaluation.

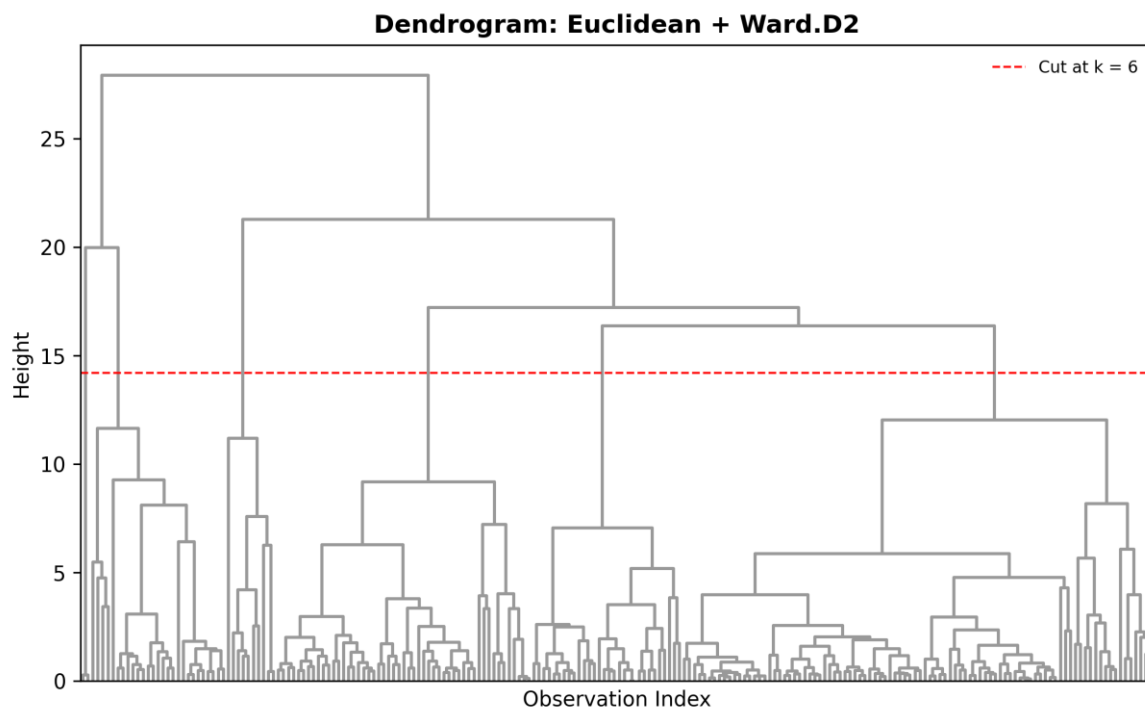


Figure 1: Dendrogram: Euclidean distance with Ward.D2 linkage. The red dashed line marks the cut at $k = 6$.

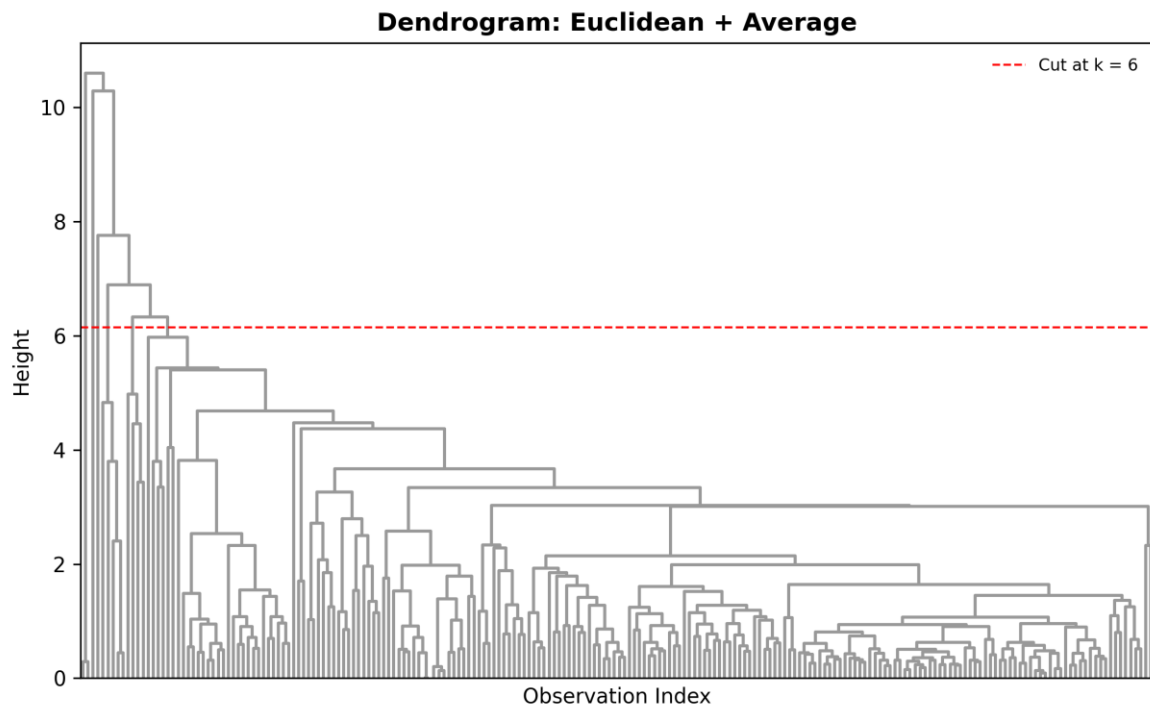


Figure 2: Dendrogram: Euclidean distance with Average linkage. The red dashed line marks the cut at $k = 6$.

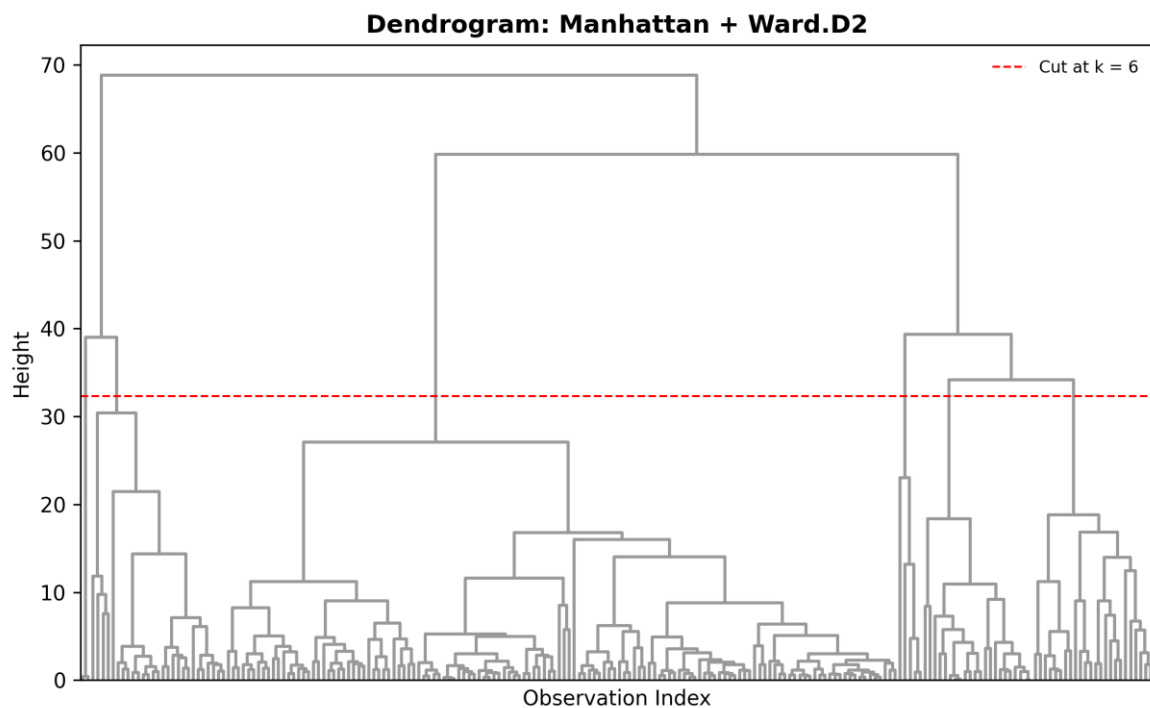


Figure 3: Dendrogram: Manhattan distance with Ward.D2 linkage. The red dashed line marks the cut at $k = 6$.

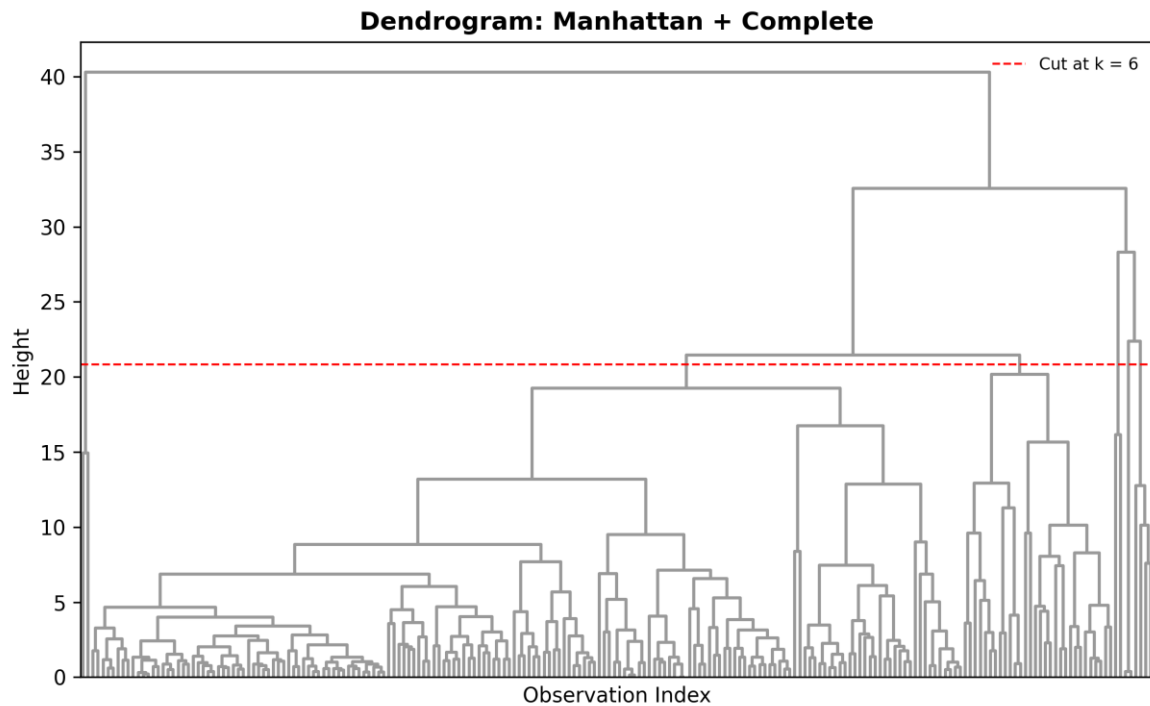


Figure 4: Dendrogram: Manhattan distance with Complete linkage. The red dashed line marks the cut at $k = 6$.

3.3 Hierarchical clustering results

After cutting each tree at the chosen number of clusters, the Hungarian method was used to find the best one-to-one matching between cluster labels and true Type labels. Table 2 reports the Hungarian error rate and R-squared value for each hierarchical combination.

Table 2: Hierarchical clustering results at $k = 6$

Label	Combination	Hungarian ER (%)	R ²
A	Euclidean + Ward.D2	57.94	0.5733
B	Euclidean + Average	62.15	0.3400
C	Manhattan + Ward.D2	48.60	0.5707
D	Manhattan + Complete	58.41	0.3654

The results show that hierarchical clustering performs only moderately on this dataset. The best hierarchical result is combination C, Manhattan + Ward.D2, with a Hungarian error rate of 48.60% and an R-squared value of 0.5707. Euclidean + Ward.D2 is second by R-squared but has a higher error rate of 57.94%. Euclidean + Average has the weakest R-squared value, mainly because average linkage creates a very large cluster that absorbs many samples from several types.

Table 3: Confusion matrix for the best hierarchical combination (Manhattan + Ward.D2, $k = 6$). Rows represent true Type labels; columns represent the matched cluster label after Hungarian assignment.

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7
Type 1	15	55	0	0	0	0
Type 2	2	61	5	8	0	0
Type 3	3	14	0	0	0	0
Type 5	0	1	0	9	2	1

Type 6	0	1	0	7	0	1
Type 7	2	2	0	0	0	25

The confusion matrix shows that most of the difficulty comes from the overlap among Types 1, 2, and 3. Type 7 is recovered more clearly than the other classes, while Types 5 and 6 are small classes and are often partially mixed with one another or with Type 2. This indicates that the chemical measurements contain some cluster structure, but the known Type categories are not naturally separated into fully distinct groups.

4 K-means Analysis

4.1 Elbow method and selected k

K-means partitions the data into k clusters by alternately assigning each observation to its nearest centroid and recomputing centroids until convergence. The total within-cluster sum of squares decreases as k increases; the elbow curve plots WSS against k and is used to judge whether adding more clusters gives only a small additional improvement.

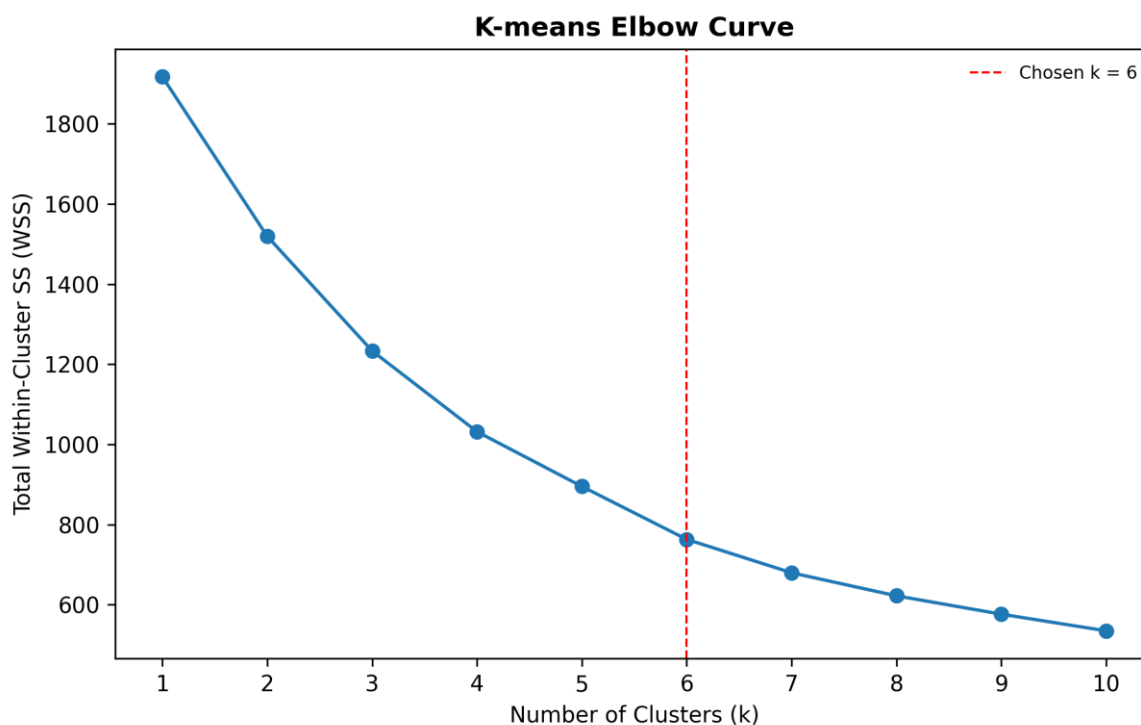


Figure 5: K-means elbow curve. The red dashed line marks $k = 6$, the number of observed glass Type categories in the dataset.

The elbow curve decreases steadily rather than showing a very sharp break. This is an important result because it suggests that the glass samples do not form perfectly separated natural groups at the observed number of classes. Nevertheless, $k = 6$ is used for the main K-means comparison because the project goal is to examine how well the six observed Type labels can be recovered from the measurement variables.

4.2 Repeated K-means runs

Because K-means depends on random initialisation, the algorithm was run 50 times for each tested k , and the solution with the lowest WSS was retained. Table 4 compares the results at $k = 5$, $k = 6$, and $k = 7$.

Table 4: K-means results for selected values of k

k	WSS	Hungarian ER (%)	R^2
5	895.19	47.66	0.5330
6	763.14	56.07	0.6019
7	679.80	54.67	0.6454

As expected, WSS decreases as k increases and R -squared increases. However, the Hungarian error rate does not improve in the same smooth way. The $k = 5$ solution has a lower error rate than $k = 6$, but it does this by collapsing at least one true Type category rather than recovering all six classes. Therefore, $k = 6$ remains the appropriate value for the main six-class recovery analysis.

4.3 PCA-based cluster view

Figure 6 shows the K-means partition at $k = 6$ projected onto the first two principal components of the standardised data. The first two components explain 27.9% and 22.8% of the standardised variance, respectively, for a combined total of 50.7%.

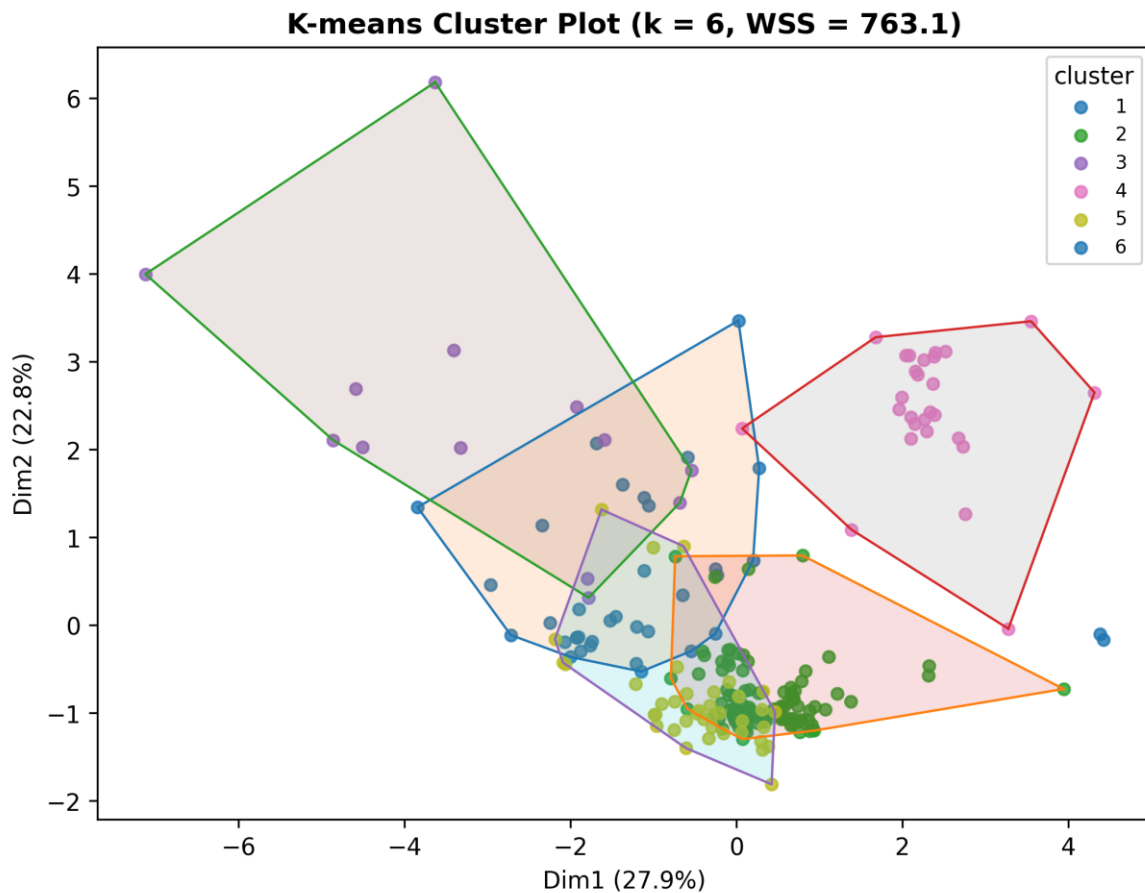


Figure 6: K-means cluster plot ($k = 6$, best of 50 runs) in the space of the first two principal components. Convex hulls enclose each cluster.

The PCA cluster plot confirms the numerical results. Some clusters are separated in the first two dimensions, but there is also substantial overlap near the centre of the plot. This overlap helps explain the high Hungarian error rate, especially for Types 1, 2, and 3.

4.4 K-means confusion matrix

Table 5: K-means confusion matrix at $k = 6$, best of 50 runs. Rows represent true Type labels; columns represent the matched cluster label after Hungarian assignment.

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7
Type 1	18	38	14	0	0	0
Type 2	6	43	20	7	0	0
Type 3	2	12	3	0	0	0
Type 5	0	2	2	6	2	1
Type 6	5	2	0	0	0	2
Type 7	3	2	0	0	0	24

The K-means confusion matrix shows that the main source of error is again the mixing of Types 1, 2, and 3. Type 7 is recovered relatively well, with 24 of its 29 observations assigned to the matched Type 7 cluster. In contrast, Type 3 has only 3 of 17 observations correctly matched, showing that this class is not separated strongly by the standardized chemical variables under the K-means geometry.

5 Final Discussion and Conclusion

5.1 Method comparison

Table 6: Final comparison of all clustering methods

Method	Hungarian ER (%)	R ²
HC: Euclidean + Ward.D2	57.94	0.5733
HC: Euclidean + Average	62.15	0.3400
HC: Manhattan + Ward.D2	48.60	0.5707
HC: Manhattan + Complete	58.41	0.3654
K-means ($k = 6$)	56.07	0.6019

5.2 Selected solution and interpretation

Table 6 shows that the best six-cluster solution is hierarchical clustering with Manhattan distance and Ward.D2 linkage. It has the lowest Hungarian error rate among the evaluated $k = 6$ methods, 48.60%, with an R-squared value of 0.5707. K-means at $k = 6$ has the highest R-squared value, 0.6019, but its Hungarian error rate is higher at 56.07%.

This difference illustrates the distinction between internal compactness and agreement with the known class labels. R-squared measures how much standardized variation is explained by the clusters, so it rewards tight geometric partitions. The Hungarian error rate measures how well those partitions recover the actual Type labels. In this dataset, a geometrically compact partition does not necessarily correspond to the official glass Type categories.

The overall conclusion is that the Glass Identification dataset is challenging to cluster. The best method is Manhattan + Ward.D2 hierarchical clustering, but even this method has a high error rate because Types 1, 2, and 3 overlap heavily and the small classes Types 5 and 6 are not consistently isolated. The clearest structure is associated with Type 7, which is recovered relatively well by both the best hierarchical method and K-means.

Appendix A: Complete Confusion Matrices

Rows represent the true Type label; columns represent the cluster label assigned by the Hungarian method. The class error rate is the number of observations from that true class assigned outside its matched cluster divided by the class size. The overall error rate is the total number of misclassified observations divided by the total sample size.

Table 7: HC confusion matrix: Euclidean + Ward.D2 (n = 214)

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7	Class ER
Type 1	20	32	18	0	0	0	50/70 (71.43%)
Type 2	4	38	26	0	8	0	38/76 (50.00%)
Type 3	4	8	5	0	0	0	12/17 (70.59%)
Type 5	0	6	2	2	2	1	11/13 (84.62%)
Type 6	0	8	0	0	0	1	9/9 (100.00%)
Type 7	2	2	0	0	0	25	4/29 (13.79%)
Overall Error Rate							124/214 (57.94%)

Table 8: HC confusion matrix: Euclidean + Average (n = 214)

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7	Class ER
Type 1	70	0	0	0	0	0	0/70 (0.00%)
Type 2	70	5	1	0	0	0	71/76 (93.42%)
Type 3	17	0	0	0	0	0	17/17 (100.00%)
Type 5	10	0	0	2	0	1	11/13 (84.62%)
Type 6	8	0	0	0	1	0	8/9 (88.89%)
Type 7	26	0	0	0	0	3	26/29 (89.66%)
Overall Error Rate							133/214 (62.15%)

Table 9: HC confusion matrix: Manhattan + Ward.D2 (n = 214)

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7	Class ER
Type 1	15	55	0	0	0	0	55/70 (78.57%)
Type 2	2	61	5	8	0	0	15/76 (19.74%)
Type 3	3	14	0	0	0	0	17/17 (100.00%)
Type 5	0	1	0	9	2	1	4/13 (30.77%)
Type 6	0	1	0	7	0	1	9/9 (100.00%)
Type 7	2	2	0	0	0	25	4/29 (13.79%)
Overall Error							104/214

Rate							(48.60%)
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Table 10: HC confusion matrix: Manhattan + Complete ($n = 214$)

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7	Class ER
Type 1	70	0	0	0	0	0	0/70 (0.00%)
Type 2	61	13	2	0	0	0	63/76 (82.89%)
Type 3	17	0	0	0	0	0	17/17 (100.00%)
Type 5	0	10	0	2	0	1	11/13 (84.62%)
Type 6	1	7	0	0	1	0	8/9 (88.89%)
Type 7	25	0	0	0	1	3	26/29 (89.66%)
Overall Error Rate							125/214 (58.41%)

Table 11: K-means confusion matrix: $k = 6$, best of 50 runs ($n = 214$)

True \ Cluster	Type 1	Type 2	Type 3	Type 5	Type 6	Type 7	Class ER
Type 1	18	38	14	0	0	0	52/70 (74.29%)
Type 2	6	43	20	7	0	0	33/76 (43.42%)
Type 3	2	12	3	0	0	0	14/17 (82.35%)
Type 5	0	2	2	6	2	1	7/13 (53.85%)
Type 6	5	2	0	0	0	2	9/9 (100.00%)
Type 7	3	2	0	0	0	24	5/29 (17.24%)
Overall Error Rate							120/214 (56.07%)

Appendix B: Use of Artificial Intelligence

ChatGPT was used as a supplementary learning resource while completing this project. Its role was limited to helping me understand the main stages of the analysis, including hierarchical clustering, K-means clustering, cluster-label matching using the Hungarian method, and how to interpret R-squared values and within-cluster sum of squares. The statistical work itself was carried out separately and was not replaced by AI. The decisions, explanations, and final conclusions in the report come from the analysis performed on the chosen Glass Identification dataset. The related R script provides the code required to recreate the report's visualizations.